NovaTech Industries — Supplier Evaluation Policy Decision Register

Checkpoint: 2.5 | **Document ID:** NT-DATA-025A | **Version:** 1.1

Date: 21 October 2026 | **Status:** Partial approval; remaining policies proposed

Scope: Supplier Recommendation Engine (FR-005)

Case study: Fictional organization and synthetic data

1. Purpose

This register records policy decisions for the supplier recommendation engine and separates confirmed decisions from recommendations requiring business approval. It supplements data-policy-and-synthetic-dataset-design.md (v1.0), rather than silently replacing it.

2. Confirmed decisions

Policy	Decision	Status	Consequence
Evaluation method	Eligibility screening followed by weighted scoring: cost 40%, delivery 25%, quality 20%, reliability 15%	Confirmed	Save weights and scoring version with each evaluation
DP-006 — Insufficient supplier history	Require manual Procurement Manager review; do not fabricate quality or reliability scores	Approved by project owner	Supplier is marked REVIEW_REQUIRED; no automatic final recommendation or purchase order
Purchase-order control	Human approval before issuing a purchase order	Previously established requirement; implementation detail pending	Scoring produces advice, not a binding purchase

A supplier with insufficient history may be shown as an *unscored candidate* for manual review; it must not be represented as a scored winner. The manager may request further evidence or authorize a documented exception. An exception must not be interpreted as a fabricated performance score.

3. Remaining decisions for approval

ID	Topic	Recommended development baseline	Status
DP-001	Comparison currency	EUR; retain original currency, exchange rate, rate source and date	Proposed
DP-002	Comparable cost	Unit price × requested quantity + attributable freight + duties + nonrecoverable fees; avoid double-counting	Proposed
DP-003	Historical lookback	Rolling 12 months as of evaluation date	Proposed
DP-004	Quality sample	At least 100 inspected units for automated quality scoring	Proposed
DP-005	Reliability sample	At least 10 completed deliveries for automated reliability scoring	Proposed
DP-007	Same-day delivery	Allow zero days; score 100 for zero-day offers, 0 for positive-day offers when fastest is zero	Proposed; revisit steep discontinuity
DP-008	Score ties	Lower comparable cost, then shorter lead time, then quotation ID	Proposed

ID	Topic	Recommended development baseline	Status
DP-009	Weights	Nonnegative and total exactly 100%; version and archive configuration	Proposed
DP-010	Approval	No automatic order issuance without human authorization	Proposed implementation policy
DP-011	Missing mandatory quotation data	Block automated ranking for that quotation and log reason	Proposed
DP-012	Missing currency rate	Block automated ranking until a valid dated rate is available	Proposed

4. Additional implementation clarifications

• **Eligibility versus manual review:** Mandatory failures (INELIGIBLE) remain distinct from missing-history cases (REVIEW_REQUIRED). Only ELIGIBLE quotations with complete evidence receive an automatic final score.

• **Sample counts:** Thresholds are based on completed deliveries and inspected units, not merely supplier age. A 12-month window with fewer observations triggers review.

• **Price precision:** Store currency amounts using decimal arithmetic; avoid floating-point rounding for financial values.

• **Quotation linkage:** Each quotation must be associated with a purchase requisition or equivalent evaluation context. Without this link, quantity and deadline constraints cannot be validated.

• **Evaluation reproducibility:** Persist comparison-set IDs, inputs, rate snapshots, weights, policy version, and evaluation timestamp.

• **No simulated approvals:** Synthetic records may represent approval outcomes, but test data must be explicitly marked as synthetic.

5. Proposed decision package

The recommended starting point is to adopt DP-001 through DP-005 and DP-008 through DP-012 as listed above. DP-007 deserves a separate review because its zero-day rule can sharply penalize next-day delivery. The development team can initially generate positive lead times (1–45 days), preserving same-day cases as a dedicated edge-case fixture until the scoring policy is approved.

Approval request: Confirm the recommended baseline for DP-001–DP-005 and DP-008–DP-012, and either accept the DP-007 zero-day rule or defer it to a later checkpoint.

6. Decision log

Date	Decision	Decision maker	Status
21 October 2026	Use weighted supplier evaluation	Project owner	Confirmed
21 October 2026	Manual review for insufficient history	Project owner	Approved
21 October 2026	Other DP policies listed above	Pending	Awaiting confirmation

7. Next action

Begin Checkpoint 2.6 with a limited, policy-neutral task: create the directory structure and three foundational reference datasets (suppliers, products, supplier_products). Do not implement automated quotation scoring until the remaining policies are approved.